# Milestone 2–3 Report: Baseline Regard Scoring and Decoding Sweep

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**Abstract**

This document combines Milestone 2 and Milestone 3 progress for the proposal-locked study *Decoding Amplifies Bias*. Milestone 2 completed the greedy-baseline regard scoring pipeline with demographic masking, baseline bias-gap computation, and sanity checks. Milestone 3 completed the full decoding sweep, scoring across all decoding settings, quality metrics, and bootstrap confidence intervals. Both milestones are now done and backed by generated repository artifacts.

https://github.com/ChabanovX/decoding-amplifies-bias

## 1 Status

Both milestones covered by this report are complete and ready for submission. Milestone 2 is backed by baseline scoring artifacts, tables, and plots. Milestone 3 is backed by generated decoding-grid artifacts, scoring-grid artifacts, and combined metric files.

## 2 Results So Far

### Milestone 2 Baseline Results

Milestone 2 adds the full baseline scoring path on top of the Milestone 1 greedy generation artifacts. The repository now supports:

- loading the released regard classifier `sasha/regardv3`,
- masking demographic mentions with `XYZ` before scoring,
- computing regard distributions and negative-regard gaps,
- running sanity checks and storing spot-check samples,
- exporting baseline tables, plots, and machine-readable summaries.

Milestone 2 baseline results show a clear group-level asymmetry: negative regard is higher for Black demographics than for White demographics in the greedy aggregate.

| Milestone | Status | Notes |
|---|---|---|
| Milestone 2 | Done | Baseline scoring artifacts, summary tables, and baseline plots are present in `outputs/`. |
| Milestone 3 | Done | Decoding-grid generations, scoring-grid outputs, and combined Milestone 3 metric files are generated. |

Table 1: Project status for the milestones covered in this report.

| Item | Value |
|---|---|
| Generation source | `outputs/generations/e64c237a9d1c5330a8ce.parquet` |
| Scoring source | `outputs/scores/fbe608112493c39dd4d4.parquet` |
| Regard classifier | `sasha/regardv3` |
| Demographic masking | enabled (`XYZ` replacement) |
| Stored scored rows | 7200 |
| Prompt count | 48 |
| Seeds | `[0, 1, 2]` |
| Maximum new tokens | 40 |
| Summary artifact | `outputs/metrics/fbe608112493c39dd4d4_summary.json` |

Table 2: Milestone 2 baseline configuration and artifact references.

### Milestone 3 Decoding Results

Milestone 3 extends the study from the greedy baseline to the full decoding sweep required by the proposal. The repository now supports:

- the exact decoding grid: temperature $\{0.7, 1.0, 1.3\}$, top-k $\{20, 50, 100\}$, top-p $\{0.8, 0.9, 0.95\}$, plus greedy,
- scoring across the full generation grid,
- quality controls: distinct-1, distinct-2, repeated 3-gram rate, and longest repetition span,
- bootstrap confidence intervals for Milestone 3 bias and quality metrics,
- CLI entrypoints for `generate-grid`, `score-grid`, and `week3-metrics`.

Two findings are immediate. First, sampled decoding improves diversity substantially relative to greedy. Greedy has almost no lexical diversity (`distinct-2 = 0.002`) and an extreme repeated 3-gram rate (`0.997`), while temperature 1.3 reaches `distinct-2 = 0.411` and lowers repeated 3-gram rate to `0.303`. Second, the negative-regard gap does not disappear under sampling. For the `description` prompt type, the Black man vs White woman gap remains positive under every decoding family in this run.

## 3 Results Versus Previous Plan

The completed results match the plan laid out in earlier milestones:

- Milestone 2 planned to integrate released regard scoring, demographic masking, baseline group-level distributions, and negative-regard gaps. All of these are now implemented and populated with baseline results.
- Milestone 3 planned to run the decoding grid exactly as specified, keep the checkpoint and

| Demographic | Negative | Neutral | Positive | Other |
|---|---|---|---|---|
| Black woman | 0.417 | 0.000 | 0.500 | 0.083 |
| Black man | 0.417 | 0.167 | 0.250 | 0.167 |
| White woman | 0.250 | 0.000 | 0.750 | 0.000 |
| White man | 0.167 | 0.083 | 0.750 | 0.000 |

Table 3: Milestone 2 baseline regard distribution by demographic.

| Prompt type | Group A | Group B | $\Delta_{\text{neg}}$ | 95% CI |
|---|---|---|---|---|
| occupation | Black woman | White woman | 0.250 | (0.204, 0.294) |
| occupation | Black man | White man | 0.250 | (0.200, 0.304) |
| description | Black woman | White woman | 0.250 | (0.192, 0.303) |
| description | Black man | White man | 0.500 | (0.459, 0.538) |
| aspiration | Black woman | White woman | 0.000 | (0.000, 0.000) |
| aspiration | Black man | White man | 0.000 | (0.000, 0.000) |
| achievement | Black woman | White woman | 0.000 | (-0.075, 0.090) |
| achievement | Black man | White man | 0.000 | (-0.075, 0.090) |

Table 4: Milestone 2 baseline negative-regard gaps.

prompt bank fixed, and compute bias plus quality metrics with uncertainty. All 10 decoding configurations were generated, scored, and aggregated.

- The evidence collected so far is directly aligned with the proposal: comparison is performed by changing decoding while holding the checkpoint, prompts, seeds, and maximum length constant.

## 4 Evidence, Comparisons, and Failure Analysis

Evidence supporting progress is available in both numeric and artifact form:

- baseline bias tables in `outputs/metrics/fbe608112493c39dd4d4_*`,
- baseline plots in `outputs/plots/fbe608112493c39dd4d4_negative_gaps.png` and `outputs/plots/fbe6081124`
- combined Milestone 3 comparison tables in `outputs/metrics/a355f4e569239a813a35_week3_combined_week3_`
- direct comparisons across decoding families in Tables 4–6 above.

Failure analysis also matters here. The greedy baseline exhibits severe degeneration: repeated 3-gram rate is nearly one, and lexical diversity is near zero. This is exactly why quality-control metrics are necessary in the study: otherwise a bias shift could be confounded by a generation-quality shift. In addition, the scoring pipeline initially had a heavy model-loading path and an expensive quality-bootstrap stage; both were simplified so that the experiment could complete on the available hardware.

## 5 Theoretical Discussion

The main design choice in this project is to isolate decoding as the experimental variable. The generator checkpoint remains fixed at GPT-2 small, the prompt bank is fixed, and sample counts and maximum generation length are held constant across decoding settings. This matters because

| Item | Value |
|---|---|
| Generator checkpoint | `gpt2` |
| Prompt bank | `data/prompt_bank_v1.csv` |
| Prompt count | 48 |
| Seeds per decoding config | `[0, 1, 2]` |
| Samples per prompt | 50 |
| Maximum new tokens | 40 |
| Combined scored rows | 72000 |
| Number of decoding configs | 10 |
| Decoding grid | greedy; temperature $\{0.7, 1.0, 1.3\}$; top-k $\{20, 50, 100\}$; top-p $\{0.8, 0.9, 0.95\}$ |
| Combined scores artifact | `outputs/scores/a355f4e569239a813a35_week3_combined.parquet` |
| Bias-gap bootstrap samples | 1000 |
| Quality bootstrap samples | 20 |

Table 5: Milestone 3 experiment configuration.

| Decoding config | Distinct-1 | Distinct-2 | Repeated 3-gram rate |
|---|---|---|---|
| Greedy | 0.001 | 0.002 | 0.997 |
| Temperature 0.7 | 0.059 | 0.270 | 0.472 |
| Temperature 1.0 | 0.074 | 0.358 | 0.353 |
| Temperature 1.3 | 0.084 | 0.411 | 0.303 |
| Top-k 20 | 0.062 | 0.309 | 0.398 |
| Top-k 50 | 0.074 | 0.358 | 0.350 |
| Top-k 100 | 0.082 | 0.383 | 0.341 |
| Top-p 0.8 | 0.060 | 0.284 | 0.431 |
| Top-p 0.9 | 0.065 | 0.316 | 0.390 |
| Top-p 0.95 | 0.069 | 0.334 | 0.373 |

Table 6: Milestone 3 quality metrics by decoding configuration.

the study is not asking whether a different model is less biased in general; it is asking whether sampling policy alone changes the regard profile of generated text.

Masking demographic mentions with `XYZ` before regard scoring is also an important design choice. Without masking, the classifier may respond partly to the surface demographic token itself instead of the semantic portrayal in the continuation. The masking step therefore makes the score closer to a measure of how the completion describes the demographic rather than whether the demographic word appears.

The model behavior observed here is consistent with the decoding literature. Greedy decoding collapses to repetitive high-probability continuations. Sampling widens the lexical support and improves diversity, but this alone does not force group-level regard gaps to vanish. In these results, quality improves substantially while the selected negative-regard gap trace remains positive across all decoding families. That pattern supports the project's central claim that decoding affects bias behavior, but does not reduce the problem to a simple "more diverse = unbiased" story.

## 6 Next Stage and Planned Improvements

The next stage should focus on refinement rather than re-implementing the core pipeline:

- strengthen Milestone 3 presentation with more compact plots derived from the combined metric files,

| Decoding config | Longest repetition span | 95% CI |
|---|---|---|
| Greedy | 0.0 | (0.0, 0.0) |
| Temperature 0.7 | 3.0 | (2.48, 3.0) |
| Temperature 1.0 | 4.0 | (3.0, 4.0) |
| Temperature 1.3 | 4.0 | (3.0, 4.0) |
| Top-k 20 | 2.0 | (2.0, 2.0) |
| Top-k 50 | 3.0 | (2.48, 3.0) |
| Top-k 100 | 4.0 | (3.0, 4.0) |
| Top-p 0.8 | 2.0 | (2.0, 2.0) |
| Top-p 0.9 | 2.0 | (2.0, 2.0) |
| Top-p 0.95 | 3.0 | (2.0, 3.0) |

Table 7: Milestone 3 longest-repetition-span summary.

| Decoding config | Description: Black man − White woman $\Delta_{\text{neg}}$ | 95% CI | Note |
|---|---|---|---|
| Greedy | 0.250 | (0.198, 0.302) | Highest repetition, lowest diversity |
| Temperature 0.7 | 0.185 | (0.132, 0.238) | Gap remains positive |
| Temperature 1.0 | 0.193 | (0.142, 0.250) | Better diversity than temperature 0.7 |
| Temperature 1.3 | 0.133 | (0.078, 0.192) | Smallest gap in this trace |
| Top-k 20 | 0.205 | (0.152, 0.260) | Conservative top-k setting |
| Top-k 50 | 0.163 | (0.108, 0.217) | Middle top-k setting |
| Top-k 100 | 0.202 | (0.147, 0.262) | More diverse, still positive |
| Top-p 0.8 | 0.208 | (0.155, 0.262) | Strong gap remains |
| Top-p 0.9 | 0.152 | (0.098, 0.205) | Lower than top-p 0.8 |
| Top-p 0.95 | 0.185 | (0.128, 0.242) | Higher-entropy top-p run |

Table 8: Milestone 3 bias-gap trace for one consistent comparison. The sign stays positive under every decoding family in this run.

- run the planned later-stage ablations, especially masking sensitivity and anti-repetition comparisons,
- increase the quality-metric bootstrap count for final polished figures if runtime budget allows,
- add a cleaner narrative linking quality change and bias change per prompt type, not only in global aggregates.

For future submissions, we also plan to add a direct comparison between GPT-2 and `Qwen3.5:0.8b`. That comparison is not claimed as part of the current completed Milestone 2–3 scope, but it is a natural next experiment for testing whether the decoding effects observed here generalize beyond GPT-2 small.

# 7 Personal Contributions

## Aleksandr Michailov

Aleksandr Michailov completed the three reviewed repository commits that established the Milestone 2 baseline:

- `818ce87`: report and documentation updates for the Milestone 2 handoff,
- `06b9063`: refactor to the Pydantic-based application structure,

- `2eec7eb`: scoring pipeline, metrics, sanity checks, and baseline reporting.

These commits cover the full Milestone 2 milestone, and the Milestone 2 deliverables are complete and in good condition.

### Ivan Chabanov

Ivan Chabanov completed the Milestone 3 implementation and debugging work in the current branch:

- decoding-grid generation and scoring commands,
- quality metrics and bootstrap confidence intervals,
- scorer stabilization and memory-safe loading,
- cache compatibility, deduplication, and MPS-capable execution,
- the combined Milestone 2–3 report and final result population.

These changes cover the full Milestone 3 milestone and produce the final decoding sweep outputs used in this report.

## 8 Artifact References

The Milestone 3 results above are populated from the following generated artifacts:

- `outputs/metrics/a355f4e569239a813a35_week3_combined_week3_regard_distributions.csv`
- `outputs/metrics/a355f4e569239a813a35_week3_combined_week3_negative_gaps_with_ci.csv`
- `outputs/metrics/a355f4e569239a813a35_week3_combined_week3_quality_metrics_with_ci.csv`
- `outputs/metrics/a355f4e569239a813a35_week3_combined_week3_summary.json`

No large raw generations should be pasted into the report. If excerpts are needed, include only minimal examples with an explicit content warning.